

FACTORY OS

Complete end-to-end costing, production readiness plan and proposed quotation

Scope locked

20	15	26	12	2	1
named users	concurrent	days/month	hours/day	locations	legal entity

Recommended founding-client commercial: ☐10L implementation + ☐1.50L/month, GST extra, for a 24-month value of ☐46.00L. Fund approximately ☐14.06L cash before official launch and carry about ☐1.07L/month while only one client exists.

This model deliberately does not use the numbers in the supplied Google Sheet. It borrows only the useful structure—identify every cost, add margin, and convert that into a quotation. It also supersedes the earlier two-page PDF, which materially undercounted production AI, database reliability, support continuity and the economics of a solo founder.

Core conclusion

The cheapest cloud bill is not the cost of delivering Factory OS. The complete cost has five layers: client-specific runtime; shared production services; Claude/Codex development capacity; one-time official-launch assurance; and founder time/support. Healthy margins come from reusing one codebase across several isolated client deployments—not from assuming credits never run out or pricing the founder at zero.

1. What is actually being costed

Factory OS is a React/Vite ERP-style production planning application with Vercel serverless APIs, Postgres/Neon persistence and Claude-powered document reading/copilot features. The planning engine itself should remain deterministic; AI reads documents and explains results.

Recommended deployment boundary

Layer	1–5 clients	Why	Later trigger
Code	One shared repository and release line	Every security/performance fix reaches every client	Never create a client fork without an explicit product decision
Application	Separate Vercel project per client, under one team	Deployment and environment isolation	Automate provisioning after repeatable product fit
Database	Separate Neon project/database per client	Clear backup, export and blast-radius boundary	Evaluate true multi-tenancy only after mature controls
Two locations	One client database with mandatory site ID	Stock, capacity, WIP and dispatch cannot mix silently	Separate only if the client requires legal/operational isolation

Usage estimate from the current product

Driver	Expected/month	Guardrail	Implication
Active concurrent user-hours	4,680	9,360	15 concurrent × 12 hours × 26 days
Function invocations	≈697,840	≈1.42M	Current 60-second orders/dispatch polling dominates
Response transfer after file fix	≈161 GB	≈670 GB	Inside Vercel allowance only if list responses stay lean
Stored files after 24 months	≈17 GB	≈34 GB	Includes 30% versioning/metadata headroom
Database after 24 months	≈11.6 GB	≈23.2 GB	Structured records only; no base64 document bodies

Critical present-state issue: PI attachments and catalogue images can live as base64 inside database JSON, while list endpoints can return that JSON repeatedly. This can hit Vercel's 4.5 MB function request/response limit and magnify bandwidth. Externalizing files, signed URLs, pagination and lean list records are launch work—not optional optimization.

2. Monthly runtime and capacity reserve

Expected cost is the modeled consumption. The guardrail is the funded subscription/credit/usage capacity that should be available so a busy month does not stop the ERP.

Cost head	Expected US\$	Funded guardrail US\$	Allocation	What it buys
Vercel Pro	20	40	Shared	Hosting, CDN, functions, WAF and overage room
Monitoring + incident response	64	64	Shared	Telemetry, uptime and on-call workflow
Transactional email	20	20	Shared	Password resets, alerts and reports
Security scanning	20	30	Shared	Dependencies, secrets and application checks
Neon Scale	55	100	Per client	Warm production, staging, 30-day PITR and enterprise controls
Anthropic production API	150	500	Per client	Document reads, copilot, retries and funded reserve
Object storage	5	10	Per client	Private slips, PIs, images, imports and versions
Independent backup	5	10	Per client	Daily encrypted dump, 90-day retention, restore drill
Domain/DNS	2	2	Per client	Client endpoint allocation
SSO/SCIM reserve	0	125	Optional	Enterprise identity connection
WhatsApp/SMS reserve	0	25	Optional	Provider usage; not included by default
Production total	341	926	One client	Excludes developer subscriptions

Solo development capacity

Tool	US\$/month	Role
Claude Max 20×	200	Primary product development
ChatGPT Pro / Codex	200	Code generation, review and testing
Claude overflow credits	100	Agentic/batch overflow with cap
Codex flexible credits	100	Work beyond included usage with cap
GitHub + 1Password + misc.	≈54	Source control, secret recovery and small services
Total	654	Shared company operating cost

One client therefore needs about □1.07L/month expected cash including the shared development tool stack, currency uplift and GST. Carry roughly □1.70L/month of funded capacity. GST may be creditable; confirm treatment with the CA.

3. AI budget and no-exhaustion design

Workload	Expected calls/month	Tokens/call in / out	Expected US\$	Stress US\$
Order slip extraction	598	4,500 / 1,200	18.84	59.20
PI extraction	260	8,000 / 3,000	17.94	57.72
ERP copilot	936	25,000 / 600	78.62	252.72
Background evals/retries	100	10,000 / 1,000	4.50	18.00
With reserve	—	—	≈149.88	≈445.78

Controls to configure before production

Control	Recommended setting	Why
Anthropic organization tier	Build tier; \$1,000 monthly cap	Expected and stress usage fit without an organization-level stop
Production workspace	\$750 workspace cap; alerts at 50/75/90%	Contains anomalies while leaving headroom
Prepaid credits	Auto-reload below \$200 back to a \$500 balance; backup card	Prepaid balance exhaustion otherwise stops API calls
Per-user usage	Soft warning, admin override and usage dashboard	One noisy user cannot consume the whole client allowance
Failure path	Save to manual review queue; exponential retry; never lose the upload	Document intake continues through 429s/timeouts
Model governance	Keep deterministic planning; AI only extracts/explains	A model change cannot silently alter production calculations

The expected token estimate uses Claude Sonnet 4.6 because that is what the current code pins. Its listed pricing is \$3 per million input tokens and \$15 per million output tokens. A model migration may lower unit cost, but it should happen only after an extraction-quality regression set is in place.

4. One-time official-launch investment

Cash workstream	Budget □	What completion means
Six-month Claude/Codex/dev-tool runway	□4.22L	Subscriptions and overflow limits funded through hardening
Independent penetration test	□2.50L	Report, remediation and retest evidence
Legal, privacy, IP and DPA	□1.25L	MSA, SOW, DPA, privacy terms and IP chain
Accounting/tax/business setup	□0.50L	GST/invoicing and financial treatment confirmed
Laptop/test devices/UPS/backup internet	□1.25L	Solo-founder operational continuity tested
First-client migration/UAT/training/travel	□1.50L	Two-site reconciliation, signed UAT and training record
Disaster recovery/release rehearsal	□0.50L	Restore and rollback drill with timestamped evidence
Subtotal	□11.72L	Before uncertainty reserve
20% contingency	□2.34L	Replace with quotes as vendors/client data are locked
Total launch cash	□14.06L	Funding gate before committing go-live

Founder time is separate

The launch plan includes 840 founder hardening hours valued at □2,500/hour = □21.00L economic investment. It covers location-aware data, audit logs, object storage, MFA/SSO readiness, staging/E2E tests, observability/backups, polling reduction, migration and UAT. This is not immediate cash salary, but it must be visible when evaluating payback or outside investment.

The earlier agency/team build estimate is therefore replaced by two transparent numbers: launch cash and founder economic time. They should never be collapsed into one figure because they answer different questions—cash runway versus product investment.

5. Proposed client quotation

Line item	Founding client	Later standard	Billing
Implementation and go-live	□10.00L	□12.00L	50% start / 30% UAT / 20% go-live
Managed subscription	□1.50L/month	□1.75L/month	Quarterly in advance
Initial term	24 months	24–36 months	8% annual price revision
Scope	20 users, 2 locations, 1 entity	Same base scope	Additional scope priced separately

Included allowances

Meter	Included/month	Overage
AI document reads	1,000 slips + PIs	□25/additional document
Copilot requests	1,500	□10/additional request
Private file storage	25 GB	□100/additional GB/month
Email notifications	10,000	Provider overage + 20% service margin
WhatsApp/SMS	Not included	Provider usage + 20% service margin
Additional user	20 base users	□2,000/user/month
Additional location	2 base sites	□25,000/month + □2L setup
Additional legal entity	1 base entity	□35,000/month + scoped setup

Commercial protection

The SOW should define modules, migration volumes, acceptance tests, support hours/severity, usage metering, data ownership/export/deletion, third-party subprocessors, change requests, annual revision and liability. Do not promise a financially backed 99.9% SLA while the company has one operational person; make service credits a paid premium only after coverage exists.

6. Why the business needs multiple clients

Shared development and production services spread across clients; the Neon database, AI runtime, private files and backups remain client-specific. Support is valued at □2,500/hour using 30 hours/month for the first client plus 8 hours for each additional client.

Clients	Cash cost/client	Support h/month	True cost/client	Margin at □1.50L	Operating decision
1	□1.07L	30	□1.82L	-21.3%	Founder-led
2	□65,114	38	□1.13L	24.9%	Founder-led
3	□51,179	46	□89,512	40.3%	Founder-led
5	□40,031	62	□71,031	52.6%	Add support help
10	□31,670	102	□57,170	61.9%	Hire support/QA
25	□26,653	222	□48,853	67.4%	Not viable as solo operation

Interpretation

At one client, the □1.50L monthly price is a market-entry price and does not fully cover founder support plus the entire shared tool stack. Around five clients, modeled true cost falls to roughly □71k/client and the quote produces about 53% gross margin. Around ten, it reaches about 62%, close to the 65% steady-state target—but 102 support hours/month means staffing must already exist.

Do not create a separate codebase for each company. Client-specific code forks destroy this cost curve because fixes, testing and deployment multiply. The permitted customization mechanism should be configuration, feature flags and scoped integrations from one shared release line.

7. Launch gates and source basis

Priority	Launch gate	Evidence before go-live
P0	Files out of database JSON; lean and paginated APIs	Largest client dataset stays below limits; signed URL tests
P0	Site ID throughout stock, capacity, WIP, orders and dispatch	Two-location isolation/reconciliation tests
P0	AI credits, caps, alerts and fallback queue	Forced 429 and zero-credit drill without data loss
P0	Credential recovery and continuity runbook	Backup owner can deploy, restore and revoke access
P1	MFA and optional SSO/SCIM	Role/access tests and security review
P1	Daily independent backup and quarterly restore	Measured RPO/RTO and restored staging copy
P1	Load and end-to-end tests against live staging Postgres	15/30-concurrent-user p95 and failure-rate report
P1	Monitoring and incident workflow	Synthetic alert and escalation drill

Primary official sources

- [Vercel Pro plan](#)
- [Vercel Functions usage/pricing](#)
- [Vercel Functions limits](#)
- [Neon pricing and Scale capabilities](#)
- [Anthropic Sonnet 4.6](#)
- [Anthropic API rate limits](#)
- [Anthropic prepaid usage credits](#)
- [ChatGPT pricing](#)
- [Codex flexible usage credits](#)
- [Better Stack pricing](#)
- [Resend pricing](#)

Model boundary

Vendor prices and limits can change. The workbook holds editable assumptions, formulas and the full source list. Penetration testing, legal, travel, insurance, communications and staffing are budgetary allowances until written quotes and the client's data/security requirements are known. Currency uses £88/US\$, 3.5% card/forex uplift and 18% GST for cash planning.